

CPS Testbed Design

Web Application and Node-RED Dashboard: User Guide

*Remotely Accessible Cyber-Physical System Testbed and Open
Architecture Synchrophasor Systems*

NSU-PGCB project (EPRC/58-2018-007-01)

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1. Overview

The CPS testbed simulates how PMU measurements travel from the power grid to a control center, how the control center uses them, and what happens when the communication network is slow, lossy, or under attack. Three simulators run together, kept in step by the **HELICS** co-simulation broker:

- **Grid (GridPACK):** solves the power flow of the chosen IEEE bus system every grid step, with slowly varying load and optional faults, and applies the control center's commands.
- **Network (NS-3):** PMUs at the optimal buses send voltage and current phasors 30 times per second over their own links to a phasor data concentrator (PDC). Attacks are applied here.
- **Control center:** estimates the state of the whole grid from the PMU data, tests the data for falsification (bad data detection), and sends voltage setpoint commands to generators when a bus voltage is out of limits.

There are two browser interfaces:

- The **web application** (port 8080) runs the full testbed: any bus system from IEEE 14 to IEEE 300, many PMUs, attacks and defenses, and detailed results for every run.
- The **Node-RED dashboard** (port 1880) runs the original single-PMU IEEE 14-bus prototype described in Sections 2–6 of the technical report.

2. Opening the Web App and the Dashboard

Both services run on the testbed server and are reached from your own computer through an SSH tunnel. With MobaXterm on Windows:

1. Click **Tunneling** in the toolbar, then **New SSH tunnel**, and choose **Local port forwarding**.
2. Enter the **Forwarded port** on your computer (left), the **SSH server** address, user ubuntu and port 22 (bottom), and remote server localhost with the **Remote port** (right). Use the values in the table below.
3. Click **Save**, attach your SSH private key with the key icon, and press **Start**. The row turns green when the tunnel is running.
4. Open the address from the table in your browser and log in.

Service	Tunnel (local → remote)	Address in your browser	Login
Web application	8080 → localhost:8080	http://localhost:8080	User admin; password in ~/.cps-web-password on the server
Node-RED dashboard	1880 → localhost:1880	http://localhost:1880/dashboard/console	User admin; password in ~/.cps-nodered-password on the server
Node-RED editor	1880 → localhost:1880	http://localhost:1880	Same as the dashboard

To show the passwords, connect to the server with MobaXterm and run:

```
cat ~/.cps-web-password ; cat ~/.cps-nodered-password
```

If you run the published Docker image on your own computer instead, no tunnel is needed: start it with `docker run -d -p 1880:1880 -p 8080:8080 -e CPS_WEB_PASSWORD=<password> ghcr.io/mazharulmd/cps-testbed:latest` and open `http://localhost:8080` (see the repository README).

3. Web Application or Node-RED?

Use the web application for experiments and results; use Node-RED to demonstrate the original prototype.

	Web application (port 8080)	Node-RED dashboard (port 1880)
Purpose	The extended multi-PMU testbed	The original single-PMU prototype
Bus systems	IEEE 14, 30, 39, 57, 118, 300	IEEE 14 only
Power flow	GridPACK solves the network live every grid step (default 0.5 s)	Replays one saved GridPACK solution, one bus per time step
PMUs	Many PMUs at the optimal buses (4 to 87; redundant option up to 202), 30 frames/s each	One PMU sending one bus value per step
Network	Own link per PMU: latency, spread, jitter, loss; PDC wait window	One link: latency and loss
Control	Closed loop: commands travel over the network and change the grid	Command is printed; the grid does not change
Attacks	Simple and stealthy false data injection, delay and drop, on any bus, with start and end times	False data injection on bus 8; drop one bus
Defenses	State estimation, chi-square bad data detection, removal of falsified PMUs, redundant PMU placement	None
Results	History of all runs; 11 summary figures and 6 charts per run; any bus can be plotted; all raw data saved	Latest run only: voltage and delay charts, gauge, table
Run time	10–80 s depending on the bus system	About 15 s

4. What Happens When You Click Run

1. **Checking.** The browser sends your settings to the server, which checks every value (ranges, and that the event and target buses exist in the chosen bus system). Problems are shown under the Run button and nothing is started.
2. **Queueing.** The experiment joins the queue. Experiments run one at a time because they share the HELICS broker; the Queue panel shows its status.
3. **Preparing.** The server loads the bus system, computes the optimal PMU placement, works out which PMU channels an attack must change, draws a latency for every PMU link, and writes the configuration files for the run.
4. **Starting the co-simulation.** The HELICS broker and the three federates (grid, network, control center) start and connect.
5. **Every grid step (0.5 s):** the grid federate applies any command that has arrived, updates the loads (and a scheduled fault, if any), runs GridPACK, and publishes the true phasors that each PMU measures.
6. **Every PMU frame (33 ms at 30 fps):** each PMU samples the phasors, adds measurement noise and sends a frame over its link; an attack changes, delays or drops the frames of the targeted PMUs. The PDC collects the frames for each time stamp and releases the set when all PMUs have reported or the wait window has passed.
7. **For every measurement set:** the control center estimates all bus voltages, runs the chi-square bad data test (removing a suspicious PMU and re-estimating if needed), checks the voltage limits and, if a bus is out of limits, sends a setpoint command to the nearest generator.
8. **Commands** travel back through the NS-3 network to the generator's node and are applied by GridPACK at the next grid step, closing the loop.
9. **Scoring.** When the simulated time is over, the server compares what the control center saw with the true grid state, computes the summary figures, and the page opens the new run automatically.

Approximate run times for 10 s of simulated time: IEEE 14 and 30 about 10 s, IEEE 57 about 15 s, IEEE 118 about 25–40 s, IEEE 300 about 80 s.

5. The Web Application Page

The header shows the project, the Principal Investigator and the grant, and a status light on the right: green **Idle** · **ready**, or amber **Running** while experiments are in the queue. Below it the page has three areas (Fig. 1): the **new experiment** form on the left, the **Queue** and **Runs** list at the top right, and the **results** of the selected run below them. The footer lists the research team and links to the GitHub repository, this guide and the technical report.

The screenshot shows the web application interface for the CPS Testbed & Synchrophasor Systems. The header includes the project name, Principal Investigator (Dr. Hafiz Abdur Rahman), and grant information (EPRC/58-2018-007-01). A status light indicates the system is 'Idle · ready'. The main content area is divided into three sections: 'New experiment' (left), 'Queue' (top right), and 'Runs' (bottom right). The 'New experiment' section contains a 'Configure & run' button and a list of experiment presets: 'Baseline' (Normal operation), 'Generator fault' (AVR setpoint - 112 pu), 'Fault + simple FDI' (1 falsified channel), 'Fault + stealthy FDI' (Evades bad data test), 'Fault + delay attack' (PMU frames +100 ms), and '5% packet loss' (Every PMU link). The 'Queue' section shows 'No jobs yet. Choose a preset or set parameters, then run.' The 'Runs' section displays a table of runs with columns: RUN, SYSTEM, ATTACK, LOSS, BDD, CONTROL, VIOLATION (S), DETECTION, and DELIVERY. The selected run is 'rpt118_fdi_simple' (IEEE 118, fdi-simple, 0 loss, on BDD, on control, 0.5 violation, 10% detection, 100.0% delivery). Below the table, the 'Results' section shows the selected run's details: 'rpt118_fdi_simple · IEEE 118-bus', '32 PMUs', '30 fps', 'latency 10 ms ±50%', 'loss 0', 'fdi-simple on bus 49 (4-9 s)', 'event avr:49:112-4', 'BDD on', and 'control on'. The 'Plot bus' is set to 49, and the 'Show' button is visible.

RUN	SYSTEM	ATTACK	LOSS	BDD	CONTROL	VIOLATION (S)	DETECTION	DELIVERY
rpt114_fdi_min 20260930 19:04:52	IEEE 14	fdi-simple	0	on	off	10.5	0%	100.0%
rpt118_fdi_stealthy 20260930 19:04:28	IEEE 118	fdi-stealthy	0	on	on	0.5	0%	100.0%
rpt118_fdi_simple 20260930 19:04:03	IEEE 118	fdi-simple	0	on	on	0.5	10%	100.0%
rpt114_fdi_red 20260930 19:03:28	IEEE 14	fdi-simple	0	on	off	10.5	100%	100.0%
rpt114_fdi_min 20260930 19:03:18	IEEE 14	fdi-simple	0	on	off	10.5	0%	100.0%
rpt118_lat100	IEEE 118	none	0	on	on	0	-	100.0%

Figure 1: The web application: experiment form (left), queue and run list (top right), results of the selected run (bottom right).

6. Inputs: Setting Up an Experiment

6.1 Presets

The preset cards at the top of the form fill it in for common scenarios. They use the bus system currently selected, so choose the **Bus system** first. After clicking a preset, you can still change any field.

Preset	What it sets up
Baseline	Normal operation: no fault, no attack, no loss.
Generator fault	At $t = 4$ s the voltage regulator (AVR) of a generator in the middle of the generator list fails and its setpoint jumps to 1.12 pu, causing an over-voltage.
Fault + simple FDI	The generator fault, and from 4 s to 9 s an attacker falsifies one PMU channel so the faulted bus appears to be at 1.02 pu.
Fault + stealthy FDI	As above, but the attacker falsifies every channel connected to the bus consistently, to avoid the bad data test.
Fault + delay attack	The generator fault, and the PMUs observing that bus are delayed by 100 ms from 4 s to 9 s.
5% packet loss	Normal operation with 5% of packets lost on every PMU link.

New experiment

Configure & run

Baseline

Normal operation

Generator fault

AVR setpoint → 1.12 pu

Fault + simple FDI

1 falsified channel

Fault + stealthy FDI

Evades bad data test

Fault + delay attack

PMU frames +100 ms

5% packet loss

Every PMU link

1 Power system

Bus system

IEEE 300-bus

300 buses · 411 branches · 69 generators · 87 PMUs (minimum placement). Highest base-case voltage: bus 149 (1.0735 pu).

Duration (s)

10

Grid step (s)

0.5

Solver

GridPACK

Event

Generator AVR f

Generator bus

221

Setpoint (pu)

1.12

Event time (s)

4

2 PMUs and network

PMU placement

Minimum

Reporting rate (fps)

30

Link latency (ms)

10

Latency spread (±)

0.5

Jitter (ms)

2

Packet loss (0–1)

0

PDC wait window (ms)

20

3 Attack

Type

FDI – stealthy

Target bus

221

Fake voltage (pu)

1.02

Extra delay (ms)

100

Start (s)

4

End (s)

9

4 Control center

Bad data detection

On

Voltage control

On

V min (pu)

0.94

V max (pu)

1.08

Run name

fault_fdi_stealthy

Random seed

1

Run experiment

Figure 2: The form after choosing IEEE 300-bus and the “Fault + stealthy FDI” preset.

6.2 Power system

Field	Meaning	Default	Allowed values
Bus system	The IEEE test network to simulate. The line below it shows the number of buses, branches, generators and PMUs.	IEEE 118	14, 30, 39, 57, 118, 300
Duration (s)	Simulated time of the experiment.	10	1–60
Grid step (s)	How often GridPACK re-solves the power flow. Commands take effect at the next step.	0.5	0.1–5
Solver	GridPACK, or the built-in Newton-Raphson solver (same results; used automatically if GridPACK fails).	GridPACK	GridPACK, Built-in NR
Event	A disturbance during the run. Generator AVR fault: a generator's voltage setpoint jumps to the faulty value. Load step: a bus load changes by a percentage.	None	None, AVR fault, Load step
Generator / Load bus	Where the event happens. For an AVR fault it must be a generator bus.	Set by preset	A bus of the chosen system
Setpoint (pu) / Change (%)	The faulty generator setpoint, or the load change in percent (negative to reduce).	1.12	Any value
Event time (s)	When the event starts.	4	0 – duration

6.3 PMUs and network

Field	Meaning	Default	Allowed values
PMU placement	Minimum: fewest PMUs that make every bus observable. Redundant: every bus seen by two PMUs (about twice as many PMUs), which removes blind spots for bad data detection.	Minimum	Minimum, Redundant
Reporting rate (fps)	PMU frames per second.	30	1–120
Link latency (ms)	Mean one-way delay of the PMU links.	10	0.1–2000
Latency spread ($\pm$)	Each link's latency is drawn between $\text{mean} \times (1 - \text{spread})$ and $\text{mean} \times (1 + \text{spread})$. 0.5 means $\pm 50\%$.	0.5	0–0.95
Jitter (ms)	Extra random delay for every frame, between 0 and this value.	2	0–500
Packet loss (0–1)	Probability that a frame is lost on each link.	0	0–0.9

PDC wait window (ms)	After the first frame of a time stamp arrives, the PDC waits this long for the others before releasing the set. Frames arriving later are counted as late.	20	1–2000
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6.4 Attack

Field	Meaning	Default	Allowed values
Type	FDI – simple: falsifies the one channel that most directly measures the target bus. FDI – stealthy: falsifies all channels connected to the target bus consistently with the network model. Delay: holds back the frames of the PMUs observing the target bus. Drop: discards those frames.	None	None, FDI simple, FDI stealthy, Delay, Drop
Target bus	The bus the attacker wants to hide. Leave empty for the bus with the highest voltage.	auto	A bus of the chosen system
Fake voltage (pu)	FDI only: the voltage the attacker wants the control center to see. Empty = true value – 0.04 pu.	auto	0.5–1.5
Extra delay (ms)	Delay attack only: how much the frames are held back.	100	0–5000
Start / End (s)	When the attack is active.	4 / 9	End must be after start

6.5 Control center and run settings

Field	Meaning	Default	Allowed values
Bad data detection	Chi-square test on the state estimation residuals; on an alarm the most suspicious PMU is removed (up to three).	On	On, Off
Voltage control	Send generator setpoint commands when a bus voltage is outside the limits. Off shows what happens without control.	On	On, Off

V min / V max (pu)	Voltage limits. The control center acts when an estimated voltage is outside them by more than 0.003 pu.	0.94 / 1.08	0.5–1.0 / 1.0–1.5
Run name	Name shown in the run list and used in the results folder name.	Set by preset	Letters, digits, _ and -, up to 40
Random seed	Controls link latencies, noise and loss. The same seed and settings give the same result.	1	Any positive integer

Click **Run experiment**. The message under the button confirms that the job was queued, or explains what to correct.

7. Outputs: Reading the Results

7.1 Queue

Shows the latest jobs with their status: **queued**, **running** (with the live log of the co-simulation), **done** or **failed** (with the error). When a job finishes, the run list refreshes and its results open.

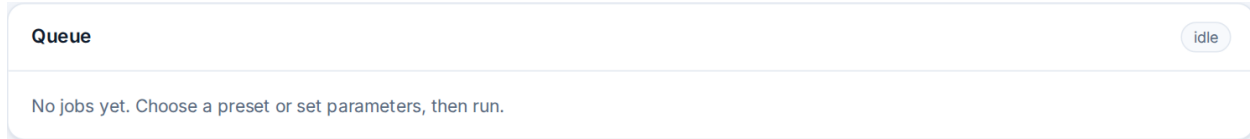


Figure 3: The queue panel when no experiment is waiting or running; the count at the top right shows active jobs.

7.2 Runs

Every finished experiment, newest first. Click a row to open its results.

Column	Meaning
Run	Run name and start time.
System	IEEE bus system.
Attack	Attack type.
Loss, BDD, Control	Packet loss rate; bad data detection on/off; voltage control on/off.
Violation (s)	How long a true voltage limit violation existed in the grid. Shown in red when longer than 1 s.
Detection	Share of measurement sets during the attack in which the bad data test raised an alarm (– when there was no attack).
Delivery	Share of PMU frames that reached the PDC in time.

7.3 Run details: summary figures

The tags under the run title repeat the main settings. The boxes below them summarise the run (Fig. 4). Their colour is a quick verdict: **green** for a good outcome (for example a violation corrected within a second, or an attack detected), **amber** for borderline and **red** for a problem (for example a long violation, an undetected attack or incomplete data); grey boxes are information only.

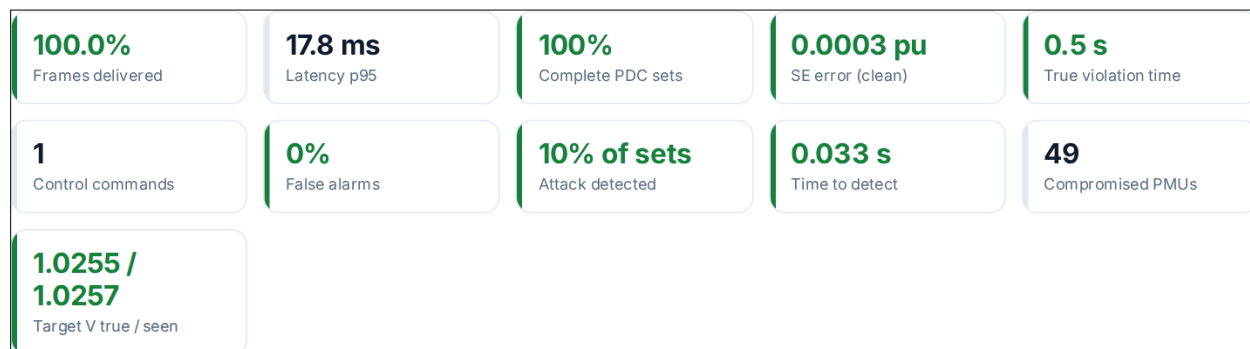


Figure 4: Summary figures of a run with a generator fault and a simple FDI attack on IEEE 118.

Figure	Meaning
Frames delivered	Share of PMU frames that reached the PDC before the wait window closed.
Latency p95	95% of delivered frames arrived within this time (PMU measurement to PDC).
Complete PDC sets	Share of measurement sets that contained every PMU.
SE error (clean)	Average difference between estimated and true bus voltages outside the attack window. About 0.0003 pu is normal.
True violation time	How long any bus voltage was really outside the limits.
Control commands	Setpoint commands applied to the grid.
False alarms	Share of sets outside the attack window with a bad data alarm. Should be about 0–1%.
Attack detected	Share of attacked sets with an alarm. An attack is usually detected only while it is hiding something, so this can be low even when detection was immediate.
Time to detect	Time from the attack start to the first alarm, or “never”.
Compromised PMUs	The PMUs whose frames the attacker changed, delayed or dropped.
Target V true / seen	Average true voltage of the target bus during the attack, and the average the control center estimated.

7.4 Run details: charts



Figure 5: The six result charts. The shaded band marks the attack window.

Chart	How to read it
Voltage at bus N	Blue: true voltage from GridPACK. Red: voltage estimated by the control center. Dashed: V max. If red leaves blue during an attack, the attack is fooling the control center. Change the bus with the Plot bus box and Show (or press Enter).
Highest bus voltage	The highest true and estimated voltage over all buses. Shows whether any violation anywhere was seen.

Bad data test (chi-square)	J is the statistic of the measurement residuals; an alarm is raised when J is above the threshold. A simple FDI makes J jump; a stealthy FDI does not. The threshold can rise briefly while suspicious PMUs are removed.
PMU → PDC latency	Histogram of the delay of delivered frames. Its spread shows why the PDC wait window matters.
Delivery per PMU	Share of each PMU's frames that arrived in time. Attacked PMUs are red.
PDC set completeness	Share of PMUs present in each measurement set over time. Loss and delay attacks make it drop.

7.5 Commands and events

The table at the bottom lists the scheduled event and every command applied to the grid: when it was issued by the control center, when it arrived over the network, when GridPACK applied it, the generator, the new setpoint, and the reason (the violating bus and its estimated voltage).

7.6 Result files

Every run is stored in its own folder, `results/runs/<date>_<time>_<name>/` (on the testbed server: `~/cps-results/`). The files can be opened with Excel, Python or MATLAB for further analysis.

File	Contents
summary.json	All summary figures of the run.
meta.json	The settings, PMU placement with every PMU's channels, and the attack plan.
run_config.json, ns3_config.json	The configuration given to the federates.
frames.csv	Every PMU frame: time stamp, PMU, send and arrival time, latency, status (delivered, late, lost, dropped), attacked.
cc_log.csv	Every measurement set: PMUs received, chi-square statistic and threshold, alarms, removed PMUs, highest and lowest estimated voltage, violations, commands.
cc_estimates.csv	Estimated voltage of every bus for every measurement set.
grid_truth.csv, grid_steps.csv	True voltage of every bus at every grid step; solver, convergence and total load per step.
commands_applied.csv	Events and commands applied to the grid.
*.log	Output of the broker and each federate, for troubleshooting.

8. Worked Examples

8.1 A generator fault with a simple false data injection (IEEE 118)

1. Choose **IEEE 118-bus**, click **Fault + simple FDI**, set the generator bus and target bus to 49, and click **Run experiment**.
2. After about 30 s the results open (Fig. 6). In **Voltage at bus 49** the true voltage jumps to 1.12 pu at 4 s and returns to about 1.015 pu at 4.5 s: the control center saw the violation and lowered the generator setpoint.
3. In **Bad data test J** jumps far above the threshold at 4 s: the falsified channel is inconsistent with the other measurements. **Time to detect** is 0.033 s (one frame) and PMU 49 was removed from the estimate, so the control center still saw the real voltage.
4. The **Commands and events** table shows the command issued at 4.067 s, delivered 40 ms later and applied at 4.5 s.

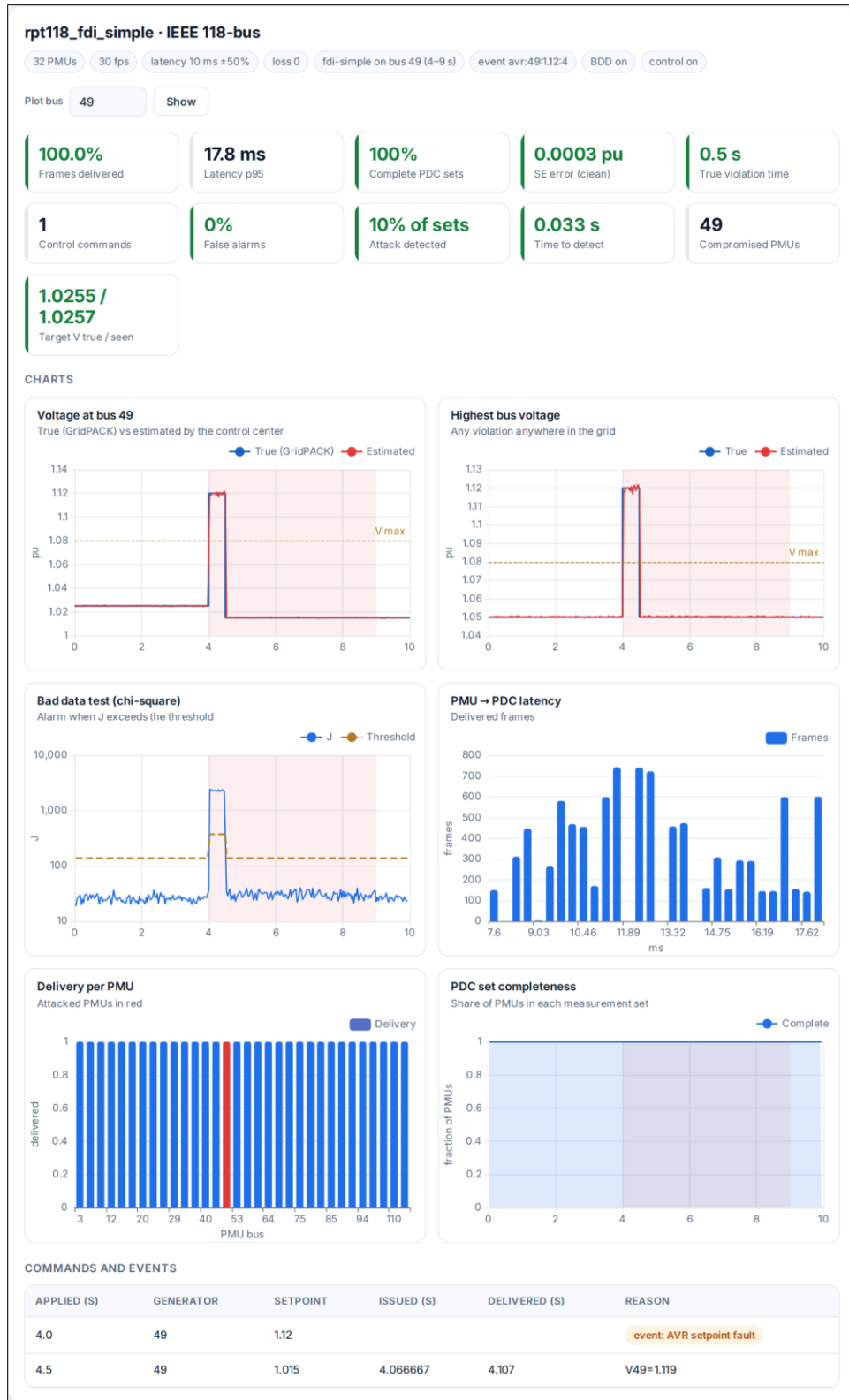


Figure 6: Results of the IEEE 118-bus generator fault with a simple FDI attack on bus 49.

8.2 The same attack made stealthy

Repeat with **Fault + stealthy FDI**. J no longer moves and the estimated voltage at bus 49 stays at the fake 1.02 pu: the bad data test is blind. The violation is nevertheless corrected, because neighbouring bus 48 also rises above the limit and the control center acts on it; plot bus 48 with the **Plot bus** box to see this.

8.3 Running the IEEE 300-bus system

1. Choose **IEEE 300-bus** in **Bus system** (300 buses, 411 branches, 69 generators, 87 PMUs).
2. Click a preset, for example **Generator fault** (it chooses generator 221), or **Baseline**.
3. Click **Run experiment**. The run takes about 80 s; the queue shows the progress.
4. Use the **Plot bus** box to plot any of the 300 buses.

8.4 Minimum versus redundant PMU placement (IEEE 14)

Run two experiments on **IEEE 14-bus** with **FDI – simple**, target bus **8**, fake voltage **1.05**, voltage control **Off**, attack 3–8 s: one with **Minimum** placement and one with **Redundant** placement. With 4 PMUs, bus 8 is observed through a single measurement, so the attack is not detected and the control center sees 1.05 pu instead of 1.09 pu. With 9 PMUs the same attack is detected in every set and the estimate stays at 1.09 pu. The **Detection** column of the run list shows 0% and 100%.

8.5 Latency and the PDC wait window

Run **Baseline** on IEEE 118 with link latency 5, 20, 50 and 100 ms. With the default 20 ms wait window, **Complete PDC sets** falls from 100% to 76%, and to 0% at 50 ms and above, and the state estimation error grows. Increasing the wait window so that it covers the spread of link latencies (for example 100 ms) should make the sets complete again, at the cost of a longer delay before each set reaches the control center.

9. The Node-RED Dashboard

The Node-RED dashboard runs the original prototype: one PMU on the IEEE 14-bus system, sending one bus voltage per time step from a saved GridPACK solution.

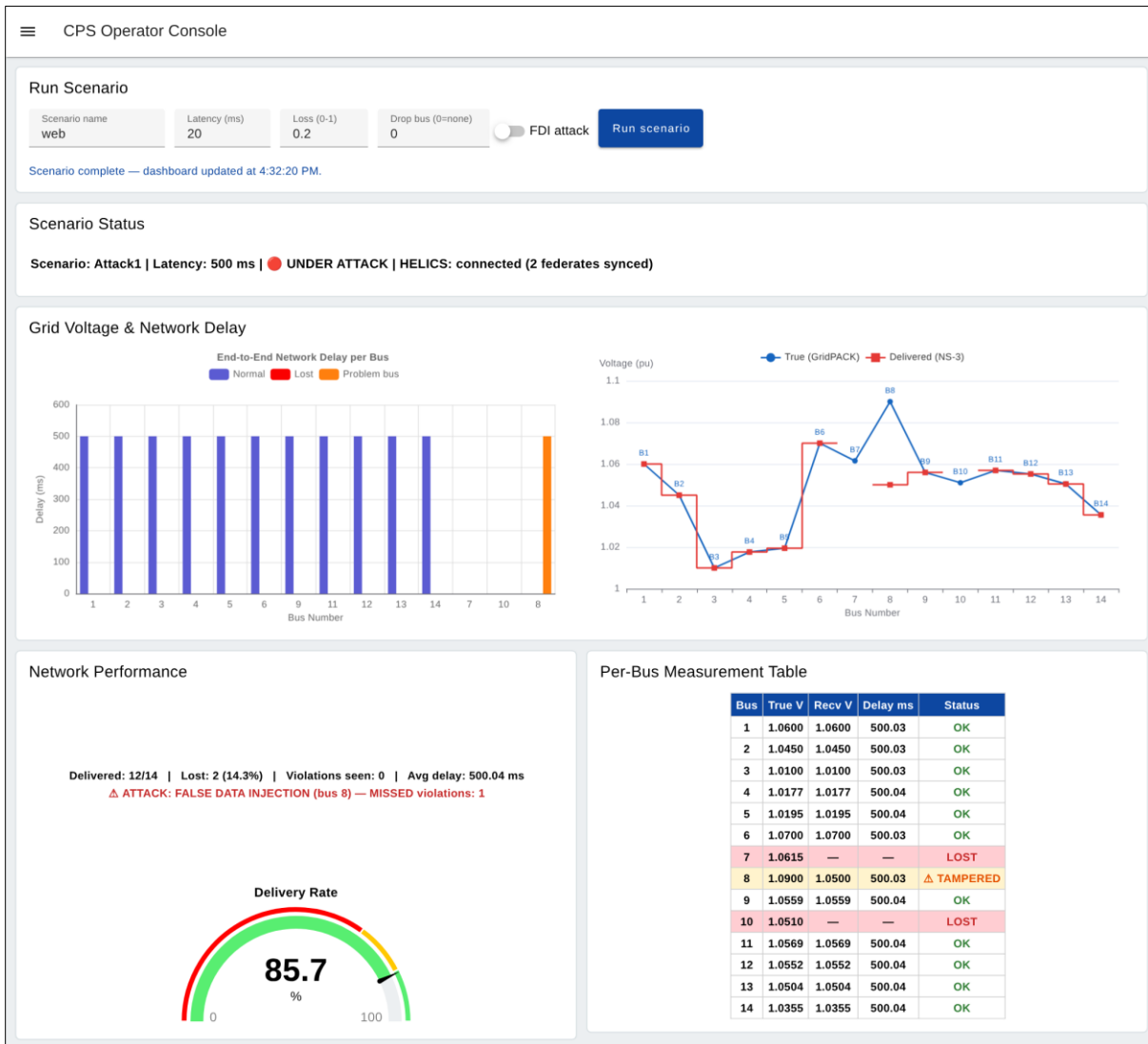


Figure 7: The Node-RED operator console after a run with an FDI attack and 500 ms latency.

9.1 Inputs (Run Scenario panel)

Field	Meaning
Scenario name	Name used for the result files.
Latency (ms)	Delay of the PMU–PDC link.
Loss (0–1)	Packet loss rate of the link.
Drop bus (0 = none)	Drop the measurement of one bus.
FDI attack	Falsify bus 8 (1.09 pu reported as 1.05 pu).
Run scenario	Starts the run; the charts update when it finishes (about 15 s).

9.2 Outputs

- **Scenario Status:** scenario name, latency, attack state and HELICS status.
- **End-to-End Network Delay per Bus:** delay of every bus's packet; lost packets and problem buses in colour.
- **Voltage chart:** true voltage (GridPACK) and delivered voltage (NS-3) for the 14 buses.
- **Network Performance:** delivered and lost packets, violations seen, missed violations, average delay, and the delivery-rate gauge.
- **Per-Bus Measurement Table:** true and received voltage, delay and status (OK, LOST, TAMPERED) of every bus.

10. Troubleshooting

Problem	What to do
The browser cannot reach <code>http://localhost:8080</code>	Start the 8080 tunnel in MobaXterm (Section 2) and check that its row is green. Try <code>http://</code> (not <code>https://</code>) or forward local port 18080 instead.
The browser asks for a login	User admin and the password from <code>~/.cps-web-password</code> (web app) or <code>~/.cps-nodered-password</code> (Node-RED).
“Could not start: bus ... is not a generator bus”	Bus numbers differ between systems. Choose the bus system before the preset, or enter a generator bus from the list in the message.
A job shows “failed”	The error is shown under the job. The federate logs (*.log) are in the run folder.
A run takes long	Runs are queued one at a time; IEEE 300 takes about 80 s. Shorter durations run faster.
Where are my results after a restart?	They are kept in <code>~/cps-results</code> on the server and appear in the run list again.
Node-RED charts do not change	Wait for “Scenario complete” under the Run button; a run takes about 15 s.